

Homework 1

CSCI 567 — Machine Learning — Fall 2026

Announced Sep 4; due Monday Sep 14, 11:59pm PT, on Gradescope

Logistics. Homework is done in pairs; submit one write-up per pair. You will submit a zip file containing the code as well as the pdf writeup. The solutions must be typed in L^AT_EX.

Collaboration and AI policy. You may discuss problems at a high level with other groups, but every pair must write its own solutions and code; do not search for solutions online, share write-ups across groups, or post course material publicly. Late days: 3 total across the semester, at most 1 per assignment (a late day is charged to each partner). AI usage is not allowed for the HWs.

Notation. As in lecture: scalars in normal font (y, γ), vectors in bold ($\mathbf{x}, \mathbf{w}$), matrices as capitals (X). The empirical risk of $\mathbf{w}$ on a sample $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ is $\hat{\mathcal{R}}(\mathbf{w})$, the true risk under the data distribution $\mathcal{D}$ is $\mathcal{R}(\mathbf{w})$, and $\mathbb{1}[\cdot]$ is the 0/1 indicator.

1 Theory-based questions

Problem 1: linear algebra and gradients warm-up (8 pts - Easy)

Each part is worth 2 pts; show your work briefly.

- 1.1** For a fixed vector $\mathbf{b} \in \mathbb{R}^n$ and a fixed *symmetric* matrix $A \in \mathbb{R}^{n \times n}$, compute the gradient with respect to $\mathbf{w}$ of

$$g(\mathbf{w}) = \mathbf{b}^\top \mathbf{w} + \frac{1}{2} \mathbf{w}^\top A \mathbf{w}.$$

- 1.2** For a nonzero $x \in \mathbb{R}^n$, what is $\text{rank}(xx^\top)$? Justify in one or two sentences.

- 1.3** For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$, show $\text{tr}(AB) = \text{tr}(BA)$.

- 1.4** Show that $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ is positive semidefinite. *Hint: write $x^\top A x$ as a sum of squares.*

Problem 2: name that failure (8 pts - Easy)

Lectures 01–02 ended with five distinct ways an evaluation number can mislead: a **generalization gap** / **sampling noise** issue, **adaptive overfitting**, a **distribution shift (external validity)** issue, a **contamination** issue, and a **construct validity** issue. For each scenario below (2 pts each), name the single best-fitting failure mode and justify it in one or two sentences.

- 2.1** A sentiment classifier is trained and tested on movie reviews collected in 2019, with excellent IID test accuracy. Deployed on 2026 social-media posts full of new slang, its accuracy drops sharply.

- 2.2** An LLM scores 95% on a well-known public benchmark. The benchmark’s test questions, with answers, have been sitting in public GitHub repositories since before the model’s training data was collected.
- 2.3** A team makes 300 successive model tweaks, keeping each tweak whenever it improves the score on the competition’s public test set. Their final public-test score turns out to be noticeably higher than their score on fresh data from the same source.
- 2.4** A model outscores physicians on a multiple-choice medical licensing exam, and a press release concludes it is ready to diagnose real patients.

Problem 3: how much can you trust a test-set number? (9 pts - Medium)

Throughout, “accuracy on a test set” is an average of i.i.d. 0/1 variables, and the model is fixed before the test set is drawn unless stated otherwise.

- 3.1 (3 pts) Sizing a test set.** How large should the test-set size n be to guarantee that the empirical test accuracy is within ϵ of true accuracy with probability at least $1 - \delta$? (Use Hoeffding’s inequality as stated in lecture.) Evaluate it at $\delta = 0.05$ for $\epsilon = 0.01$ and for $\epsilon = 0.001$, and state the general lesson about what one more decimal digit of certified precision costs.
- 3.2 (6 pts) Two test sets, two intervals.** In lecture, Hoeffding connected *one* empirical accuracy to *the* true value through a single confidence interval. The ImageNet replication (*Do ImageNet Classifiers Generalize to ImageNet?*) could not do that: there is no oracle for the true accuracy, only **two finite test sets** — so there are really **two intervals**, both trying to capture the *same* unknown true accuracy μ . A fixed model is evaluated on two independent test sets of sizes n_1 and n_2 , both drawn i.i.d. from the same $\mathcal{D}$, giving measured accuracies A_1 and A_2 .
- (i) From Hoeffding, each test set yields a confidence interval $[A_j - \epsilon_j, A_j + \epsilon_j]$ that contains μ with probability at least $1 - \delta_j$, where $\delta_j = 2e^{-2n_j\epsilon_j^2}$. Compute the two 95% half-widths for $n_1 = 50,000$ and $n_2 = 10,000$.
 - (ii) Say the two intervals are *simultaneously valid* if both contain μ . Show that (a) simultaneous validity fails with probability at most $\delta_1 + \delta_2$, and (b) whenever the intervals are simultaneously valid they must *overlap*, so $|A_1 - A_2| \leq \epsilon_1 + \epsilon_2$.

Problem 4: why mini-batches are less noisy than single-example SGD (5 pts - Easy)

Lecture 02 replaced the full gradient by the gradient of one random example, and mentioned that averaging a *mini-batch* of examples “trades some noise for steps that parallelize well.” Here you make the noise reduction precise.

Fix the current parameter $\mathbf{w}$ and write $\mathbf{g}_i \in \mathbb{R}^d$ for the gradient of the i -th example’s loss at $\mathbf{w}$, so the full gradient is $\mathbf{g} = \nabla \hat{\mathcal{R}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}_i$. Define the **gradient noise** of the training set at $\mathbf{w}$ as

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{g}\|^2.$$

Everything below is about the randomness of *which examples* are drawn; the data and $\mathbf{w}$ are fixed.

4.1 (2 pts) One example. An index I is drawn uniformly from $\{1, \dots, n\}$ and the step uses $\mathbf{g}_I$. Show that $\mathbb{E}[\mathbf{g}_I] = \mathbf{g}$ (this is the unbiasedness claim from lecture) and that $\mathbb{E}[\|\mathbf{g}_I - \mathbf{g}\|^2] = \sigma^2$.

4.2 (3 pts) A mini-batch. Now draw B indices $I_1, \dots, I_B$ independently, each uniform on $\{1, \dots, n\}$ (so an index may repeat), and use the averaged gradient $\hat{\mathbf{g}}_B = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_{I_b}$. Show that $\mathbb{E}[\hat{\mathbf{g}}_B] = \mathbf{g}$ and

$$\mathbb{E}[\|\hat{\mathbf{g}}_B - \mathbf{g}\|^2] = \frac{\sigma^2}{B}.$$

Hint: write $\hat{\mathbf{g}}_B - \mathbf{g} = \frac{1}{B} \sum_b (\mathbf{g}_{I_b} - \mathbf{g})$, expand the squared norm, and use independence to kill the cross terms.

2 Programming-based questions

Setup. Use Python 3 (≥ 3.9) with **numpy and matplotlib only** — no scikit-learn, scipy, pytorch, pandas, or other ML/numerics packages (ask first if you think you need something else). Starter code is provided in the `hw1_code` folder posted with this homework; fill in every function marked `TODO` without changing seeds, splits, or reporting code, so that reported numbers are comparable across submissions.

What to submit. Submit your completed `.py` files in a single zip file. The answers to the questions (reported numbers, plots, etc.) should be written up and submitted as part of your L^AT_EX solution.

Problem 5: linear regression — closed form, GD, SGD (20 pts - Medium)

We fit the linear model from Lecture 02 on synthetic data where we know the truth. The starter code (`linear_regression.py`) draws $n = 1000$ training points $\mathbf{x}_i \in \mathbb{R}^{100}$ i.i.d. from a standard Gaussian, picks a true coefficient vector $\mathbf{w}^* \sim \mathcal{N}(\mathbf{0}, I)$, and sets $y_i = \langle \mathbf{w}^*, \mathbf{x}_i \rangle + \varepsilon_i$ with noise $\varepsilon_i \sim \mathcal{N}(0, 0.5^2)$. The objective is the empirical risk from lecture,

$$\hat{\mathcal{R}}(\mathbf{w}) = \frac{1}{2n} \sum_{i=1}^n \left(\mathbf{w}^\top \mathbf{x}_i - y_i \right)^2.$$

5.1 (6 pts) Closed form. Implement `emp_risk` and `closed_form` (solve the normal equations $X^\top X \mathbf{w} = X^\top y$; use `np.linalg.solve`, not an explicit inverse). Report: (i) the training risk $\hat{\mathcal{R}}(\mathbf{w}_{\text{LS}})$, (ii) the training risk of the all-zeros vector, (iii) the risk of $\mathbf{w}_{\text{LS}}$ on the freshly drawn test set the starter code provides. Comment on the train–test gap, and on how the training risk compares to the noise floor $\text{Var}[\varepsilon]/2 = 0.125$ — which side of it does the training risk land on, and why?

5.2 (7 pts) Gradient descent. Derive the gradient of $\hat{\mathcal{R}}$ (one line, from lecture), implement `full_gradient` and `gradient_descent`, and run 20 iterations from $\mathbf{w} = \mathbf{0}$ with step sizes $\eta \in \{0.005, 0.5, 1.3\}$. Plot all three risk curves on one log-scale figure, report the final risk for each, and explain the three regimes you see in terms of Lecture 02’s step-size discussion. Which step size gets closest to the closed-form optimum, and how close?

5.3 (7 pts) SGD. Implement `sgd`: at each step draw one training index uniformly and step on the single-example loss $\frac{1}{2}(\mathbf{w}^\top \mathbf{x}_i - y_i)^2$. Run 1000 iterations from $\mathbf{w} = \mathbf{0}$ with $\eta \in \{0.001, 0.01, 0.02\}$; plot the three curves, report the final risks, and compare against GD: gradient evaluations per step, passes over the data used by each method in total, the shape of the curves (why is SGD's jagged?), and which method reached a lower objective here.